

EMIR BEKAR

Mechatronics Engineer · Game Developer · Content Creator

Istanbul, Turkey | emir.bekar@bilgiedu.net | emiukob.github.io | linkedin.com/in/emir-bekar-883410360 | github.com/emiukob

SUMMARY

Mechatronics Engineering graduate (100% English) with strong mathematical and analytical foundations, award-winning senior design research, and two IEEE-published papers in multi-agent robotics and AI. Hands-on experience developing mobile games in Unity and managing an active 12,000+ subscriber YouTube community with direct insight into player retention and content iteration. Passionate about mobile casual games and eager to drive engaging product features at Peak.

EDUCATION

Istanbul Bilgi University

Jul 2026

Bachelor of Science in Mechatronics Engineering (100% English Curriculum)

Istanbul, Turkey

- 🏆 **Best Senior Design Project Award:** "Adaptive Formation Control for Collaborative Swarm Robots" (Presented at ICHORA 2026, published on IEEE Xplore).

FEATURED PROJECTS

Match 3D Mobile Game

Unity 6 · C# · Blender · URP

- Built a hybrid-casual 3D matching mobile game featuring physics-based item matching and zero-GC object pooling for stable 60 FPS.
- Designed core retention and monetization meta-game loops: Daily Rewards, Lucky Spin wheel, and in-game economy store.
- Modeled and textured modular 3D assets in Blender tailored for mobile visual clarity. (github.com/emiukob/Match3D)

AC-130 Gunship Simulation (In Development)

Unity 6 · C# · Custom Shaders

- Close air support tactical simulation with custom FLIR thermal vision shaders, dynamic terrain destruction, and avionics HUD telemetry.

Interactive 8-Bit Island Map Portfolio

JavaScript · Web Audio API · CSS

- Gamified bilingual portfolio with a walkable island canvas, synthesized chiptune audio engine, and mobile-first 100dvh layout (emiukob.github.io).

Forza Horizon 6 Telemetry Cockpit & LAN Drop

Python · FastAPI · UDP · Node.js

- Wireless 60Hz real-time telemetry mobile dashboard and zero-config local network file streaming tool with 8MB chunking.

PUBLICATIONS (IEEE XPLORE · HORA CONGRESS)

- Adaptive Formation Control for Collaborative Swarm Robots (IEEE 2026):** Leader-follower formation switching via FSM and fuzzy-logic ACC control eliminating stop-and-go oscillations, validated in Webots simulation.
- Multi Agent Reinforcement Learning Based Swarm Navigation (IEEE 2026):** Autonomous swarm coordination and collision-free path optimization using Multi-Agent Reinforcement Learning (MARL).

COMMUNITY & CONTENT CREATION

Emiukob – YouTube Creator & Mod Developer

12,000+ Subscribers · youtube.com/@emiukob

- Built a gaming and tech community to 12,000+ subscribers; analyzed audience retention curves, drop-off rates, and CTR to iteratively optimize content pacing.
- Created and published "Hologram Projector" 3D hologram mod for Minecraft in Java on CurseForge.

SKILLS & CERTIFICATIONS

Game & Product: Mobile casual / hybrid-casual games, 3D match mechanics, level pacing, FTUE, player retention & meta-game loops

Technical & Tools: Unity 6 (C#), Python, JavaScript, Java, Node.js, Blender, Web Audio API, Git/GitHub, Webots Simulation

Analytical: Data interpretation, A/B testing mindset, mathematical modeling, fuzzy logic, reinforcement learning

Certifications: Basic Python for Business Analytics (Arizona State University) | UAV 0/1 Pilot License (SHGM)